

Reviewer Pack — Minimal Rank of Quasi-Projective Varieties over Boolean Func...

Entry 3baf27341104 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quasi-Projective Varieties over Boolean Functions vs Communication Complexity for DISJOINTNESS

Entry ID: 3baf27341104 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 13:59:18 UTC

1. Conjecture

Field A (mathematical branch): Quasi-Projective Geometry **Field B** (complexity object): Communication Complexity

Statement:

{'text': 'For every quasi-projective variety V defined over the finite field with two elements F_2 , there exists a correspondence between the minimal rank of V and the randomized communication complexity for the DISJOINTNESS problem. Specifically, if the minimal rank of V is r , then the randomized communication complexity of DISJOINTNESS is at least 2^r .', 'quantity_relation': ' $E[X(\text{instance})] = \Theta(2^{(\min_rank(V))})$ ', 'counterexample': 'Quasi-Projective variety V such that $2^{(\min_rank(V))}$ '}

Rationale (proposer's reasoning):

{'text': 'Quasi-projective geometry offers a rich algebraic structure that could potentially encode the complexity of communication tasks. The minimal rank of a quasi-projective variety may reflect the intrinsic difficulty of distinguishing between distinct states in the DISJOINTNESS problem, which is a fundamental challenge in communication complexity.'}

Taxonomy category: COMM_DISJ (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): d5b1b6e9dc82dea6

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for a statistically significant number of random Boolean functions and their associated quasi-projective varieties, the minimum rank of V matches the randomized communication complexity for DISJOINTNESS to within an error margin of $\pm 10\%$ with at least 100 seeds. The criterion is falsified if any seed results in a discrepancy between the minimal rank of V and the communication complexity greater than $\pm 10\%$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.80	HITS	SAFE
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def construct_quasi_projective_variety(f):
        # Simplified construction using polynomial degree
        return len(f)

    def communication_complexity(n):
        # Simulated complexity based on function size
        return n

    n = random.randint(5, 40) # Sweep through different sizes
    f = generate_boolean_function(n)
    min_rank = construct_quasi_projective_variety(f)
    comm_complexity = communication_complexity(n)

    metric_name = "communication_complexity"
    metric_value = comm_complexity
    instances_tested = 1
    conjecture_holds = abs(metric_value - 2**min_rank) <= 0.1 * 2**min_rank
    counterexample = "" if conjecture_holds else f"Discrepancy: {metric_value} != 2^{min_rank}"

    return {
        "metric_name": metric_name,
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
```

```

        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(res["metric_value"] for res in results) / len(results)
    std_value = math.sqrt(sum((res["metric_value"] - mean_value)**2 for res in results) / len(results))
    support_fraction = sum(1 for res in results if res["conjecture_holds"]) / len(results)

    if all(res["conjecture_holds"] for res in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not res["conjecture_holds"] for res in results) and support_fraction >= 0.8:
        first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Discrepancy\" first_failing_seed={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE support_fraction={support_fraction}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it did not meet the pre-registered support condition of running for at least 240 seconds and providing statistically significant results. | next: Re-run the test with increased time limits to ensure it completes and meets the pre-registered criteria.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11013
2	preregistration	ollama_remote	glm4:latest	0	0	6069
3	novelty	ollama_remote	glm4:latest	0	0	5026
4	novelty	ollama_remote	glm4:latest	0	0	5046
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13039
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9497
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7122
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7121
9	judge	ollama_remote	glm4:latest	0	0	22828

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 86762 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/3baf27341104.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/3baf27341104.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/3baf27341104.tar.gz` (if generated)